

SmartHire AI

An Agentic AI Assistant Reimagining the Recruitment
Lifecycle



RAG SOURCED



TOOL CALLING



HUMAN-IN-THE-LOOP

Team Name – Team Elite

The Recruitment Problem

Why modern recruiting is bogged down in manual,
administrative fatigue

| Recruitment Bottlenecks Today

The Administrative Burden

Recruiters expend over 60% of their bandwidth on transactional and manual tasks: sifting through hundreds of repetitive resumes, tailoring outreach messages, rewriting job descriptions, and scheduling calls.

Passive Legacy ATS

Traditional Applicant Tracking Systems (ATS) act merely as passive data vaults. They require manual triggering, lack natural language semantics, and fail to provide intelligent insights, trends, or salary data contextually.

The Solution: SmartHire AI

A conversational, multi-agent assistant designed to handle
autonomous recruiting workflows

| Core Capability Pillars



Intelligent Sourcing

Retrieves high-context, indexed candidate profile details semantically from local vector stores using advanced RAG pipelines.



Agentic Orchestration

Uses intelligent routing, multi-agent pipelines, and context-aware logic to autonomously perform complex multi-step recruitment flows.



Human-in-the-Loop

Ensures robust security and control by requiring explicit recruiter confirmation prior to execution of high-risk business triggers.

Autonomous Routing Architecture

SmartHire AI functions as a cognitive orchestrator, analyzing incoming recruiter prompts and routing them to the optimal resource:

-  **RAG / Vector Databases**
Invoked for semantic, query-based candidate screening and deep profile extraction.
-  **Deterministic Computations**
Calculates metrics, score cards, and matching values locally without prone-to-hallucination LLM logic.
-  **Live Web Queries**
Performs real-time search engine sweeps to fetch current salary ranges and competitor trends.



Semantic Resume Screening

Going Beyond Keyword Searching

Traditional parsers fail when CV keywords don't match. SmartHire AI translates entire applicant files into vector embeddings, matching skills contextually.

Using a dense vector search model, the engine ranks candidates based on experience depth, technical capabilities, and hidden potential, drastically reducing evaluation bias.

Alice Hunt

Pre-screen Candidate for Sales Account Executive

Criteria 5% Psychometrics Interview transcript Candidate log Notes

Summary

on mechanical engineering with a master's in management.

experience as a project engineer at Mammoet, handling technical calculations and project management.

ch and English with limited proficiency in German.

on strategic consulting and project management.

ertificate of Excellence for diploma thesis.

Strengths & challenges for Sales Account Executive

English and Dutch

Professional proficiency in English and native proficiency in Dutch,

Challenges

Lack of Sales Experience

Alice's CV does not demonstrate

| End-to-End Agentic Workflow

1

Semantic Screening

Retrieves matches from CV database using Vector search.

2

Live Benchmarking

Performs live web queries for competitive salary metrics.

3

Generate Questions

Drafts high-quality technical interviewing rubrics.

4

Recruiter Approval

Human reviews draft and authorizes candidate contact.

Efficiency Gains: Slashed Hours



| Recruiter Allocation Reimagined



	Relationship & Placement Strategy	55%
	Interviews & Selection	25%
	Collaborative Strategy Meetings	15%
	Admin/Data Tracking	5%

Semantic RAG Similarity Model

The matching accuracy between a candidate profile vector and the JD vector is determined by **Cosine Similarity**:

$$\text{Matching Score} = \frac{A \cdot B}{\|A\| \|B\|}$$

Trigger Scenario	Core Agent Routing	Executed Tool	Status
Compare python resumes to job brief	Semantic Context Match	RA RAG DB Tool	Auto
Lookup local DevOps salary ranges	Real-time Web Sourcing	We Web Search Engine	Auto
Send interview invites & offer drafts	High-Stakes Dispatch	Em Email Gateway	ITL Confirmed

THANK YOU !!